GOPI CHAITANYA ALAMANDA

Railway Technology Engineer | Python Automation & System Integration

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Notice Period: 3 Months | linkedin.com/in/gopi-chaitanya-alamanda | github.com/GopiChaitanyaAlamanda

PROFESSIONAL SUMMARY

Railway Technology Professional with 2.7+ years of experience at Efftronics Systems Pvt. Ltd. specializing in railway signaling application deployment, core technical support, real-time data analytics, and Python-based automation. Proven track record of optimizing mission-critical engineering workflows across 1,000+ Indian Railways stations. Strong grasp of signaling principles, interlocking concepts, and safety-critical system logic. Equipped with foundational/theoretical knowledge of modern train control systems (ETCS/ERTMS, CBTC, Kavach) and CENELEC standards to drive deployment, integration, and validation strategies for global rail tech companies.

CORE TECHNICAL & RAILWAY DOMAIN SKILLS

- **Railway Systems (Deployment & Support):** Train Monitoring Systems (TMS), Automatic Train Tracking Systems (ATTS), Remote Diagnostic and Preventive Maintenance (RDPM), Safety Point Alarm Unit (SPAU), Control Office Application (COA), Nirdesh Alerts.
- **Signaling & Interlocking Concepts:** Route Setting, Route Locking, Route Releasing & Route Cancellation; Route Relay Interlocking (RRI) & Panel Interlocking (PI) Station Operations.
- **Block Section Working Knowledge:** Absolute Block System (ABS), Automatic Signaling, Intermediate Block Huts (IBH).
- **Advanced Control & Standards (Theoretical):** CENELEC EN 50126 (RAMS), EN 50128 (Software), EN 50129 (Safety-Related Electronic Systems), SIL Levels (SIL 0–4), ETCS/ERTMS (Levels 0–3), CBTC Principles, Kavach (Indian ATP), ATO, ATS.
- **Software & Automation Engineering:** Python (Expert), Oracle SQL (Expert), Flask, Django, React JS, Azure VM, JSON Processing, Advanced Excel Automation, Pandas, NumPy, TCP/IP, IoT Monitoring.

PROFESSIONAL EXPERIENCE

Efftronics Systems Pvt. Ltd. | Engineer Technical Support

June 2023 – Present | Vijayawada, Andhra Pradesh

- Configured, deployed, and stabilized high-availability railway signaling and train tracking applications across a network of 1,000+ railway stations.
- Designed and engineered Digital Twin visualization systems for critical theater signaling circuits (Signals, Points, Track Circuits, and Hot Metal Units) to execute real-time flow and health monitoring.
- Formulated advanced Oracle SQL analytical queries (30+ lines containing complex multi-table joins and conditional CASE logic) to identify train tracking break patterns on ATTS data, directly resulting in tracking algorithm upgrades in collaboration with CRIS.
- Lead team of 5 members & built automated testing utilities to generate route-wise validation test cases for occupied platform conditions (SPAU), lowering validation cycle times across 300+ stations with zero safety incidents.
- Hand-picked to lead a team of 7 interns tasked with deploying the Nirdesh Alerts real-time failure module across 145 target stations in Jaipur and Allahabad divisions.
- Contributed to the system deployment of India’s largest railway video wall network (116 stations, inaugurated by the Railway Minister). Selected to present live technical solutions at the IREE and IRSE exhibitions in New Delhi.

- Prepared and presented rigorous Acceptance Test Procedures (ATPs) to RDSO and RITES officials to facilitate official system validation and product sign-off.

MAJOR IMPACT PROJECTS

Fault Standardization Automation Tool (Fault Replacer) *(Python, Oracle SQL)*

Problem: 18 Eastern Dedicated Freight Corridor (EDFCC) projects required manual standardization of alarm/fault names across 61 stations—a workflow estimated at 30 days of manual effort.
Solution: Built a Python processing script with intelligent before/after data mapping to execute bulk database modifications automatically.
Impact: Reduced execution time from 30 days to under 5 minutes with 100% data accuracy and zero visual config discrepancies.

ChronoTech (Database Skeleton Prep Utility) *(Python, Oracle, Excel/CSV)*

Problem: Manual configuration of database schemas from bulk data feeds required 15 minutes per 4,096 station asset records.
Solution: Programmed an automatic parsing engine that generated structural database dumps dynamically from source sheets.
Impact: Compressed processing time from 15 minutes to 2 minutes, earning an immediate corporate transition from site operations to the core engineering division.

Engineering Automation Suite *(Python, JSON, TCP/IP, File Systems)*

Developed a 4-tool workflow network used internally across multiple deployment branches:
• ESPAU Test Generator: Automated route safety test suites, scaling validation capacity across 300+ stations (90 min to 30 min per station).
• JSON Config Processor: Batch-edited asset properties, text labels, and flow orientations for Nirdesh UI dashboards instantly.
• Auto File Collector & Dump Creator: Automated localized Oracle backups and collects Application Config files for Cloud Deployment ready setups, eliminating hours of manual retrieval from nested date folders.

EDUCATION

BTech in Computer Science Engineering [AI & DS] | Kakinada Institute of Engineering and Technology
2025 – 2028 [Expected]

Diploma in Computer Science Engineering | Government Polytechnic College, Srikakulam, AP
Pursued: 2020 - 2023 | Aggregate: 80%

Secondary School Certificate [10th Class] | Government High School, Bheemunipatnam, AP
Pursued: 2020 | Aggregate: 80%

RECOGNITION & PROFESSIONAL CONTRIBUTIONS

- Director Public Recognition: Formally cited by the corporate Director during an all-hands assembly for exceptional code quality and efficiency contributions.
- Executive Appreciation: Recipient of two official management letters of appreciation for pioneering engineering automation tools.
- Industry Representations: Selected as a core corporate technical presenter at the IREE 2023 and IRSE 2023 signaling and telecommunication exhibitions in New Delhi.
- Note on Domain Scope: Technical knowledge regarding ETCS/ERTMS, CBTC, Kavach architectures, and CENELEC EN standards has been acquired via intensive independent study, internal documentation review, and technical exchange at industrial forums rather than direct site deployment.